

NISHCHAL MARUR

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SUMMARY

Machine Learning Engineer with 4+ years of experience delivering end-to-end Production ML systems. Engineered high-stakes verification pipelines for luxury goods at 96% TPR, developed real-time recommendation engines and Agentic RAG pipelines, while compressing edge models by 10x for scalable mobile deployment.

EXPERIENCE

AI Engineer Intern | Connyct Inc | New York Nov 2025 - Jan 2026

- Built a two-tower recommendation engine that achieved a Recall@10 of 0.85, delivering personalized event suggestions at <40ms latency to maximize user engagement.
- Deployed location-aware reranking and A/B testing frameworks for an LLM orchestrator with MCP tools, directly optimizing CTR and search relevance through composite scoring and Redis caching.

Research Assistant | Perception and Robotics Group | Maryland Jun 2025 - Aug 2025

- Developed a cross-modal audio generation pipeline converting performer pose sequences to violin audio using DDSP, Transformer encoders, and autoencoder-based MIDI synthesis, enabling audio generation directly from motion capture.

Machine Learning Engineer II | Entrupy Inc | Bangalore Aug 2021 - Aug 2024

- Automated 95% of manual verification volume for 150K+ luxury items monthly by building a vision-based deep learning authentication pipeline achieving 96% TPR at 5% FPR.
- Led R&D of a 3D document unwarping system using synthetic data that outperformed SOTA benchmarks, resulting in a 23% increase in OCR accuracy with 0.84 SSIM for real-world documents.
- Achieved 10x model compression for on-device deployment via EfficientNet distillation from SAM-HQ and FP16 quantization, enabling real-time auto-capture with 96% IoU on consumer iOS devices.
- Improved macro fingerprinting pipeline for luxury goods return fraud tracking using patch embedding similarity, boosting TPR by 15% and reducing latency with Ray parallel processing.
- Reduced time-to-production for new models from weeks to days by designing synthetic data pipelines using Blender and Stable Diffusion to simulate camera intrinsics, lighting, and complex material textures.

MLOps Intern | IBM | Bangalore Jan 2021 - Jul 2021

- Enhanced throughput in IBM Watson Cloud by reducing inference latency 15ms through Go-based concurrency optimization and Kubernetes batch prediction pipelines.
- Evaluated performance trade-offs across Tensorflow, PyTorch and ONNX runtimes to inform the design of a new internal inference architecture for enterprise-scale deployments.

Software Engineering Intern | SLK Software | Bangalore May 2020 - Jul 2020

- Eliminated 10+ hours of weekly debugging time by building a centralized log aggregation system using ELK Stack, Filebeat, and NodeJS APIs across distributed components.

TECHNICAL SKILLS

Languages: Python, C++, Go, SQL, Scala, Node.js, Bash

ML Frameworks: PyTorch, TensorFlow, HuggingFace, OpenCV, Scikit-Learn, SAM, CoreML, ONNX, TensorRT

GenAI & Retrieval: LangChain, LangGraph, LlamaIndex, vLLM, LoRA, Elasticsearch, MCP, FAISS, Stable Diffusion

Cloud & Infra: AWS, GCP, Docker, Kubernetes, Ray, Kubeflow, PySpark, MLflow, Triton, FastAPI, Redis, Kafka, PostgreSQL, CI/CD

ML Domains: Computer Vision, Large Language Models, MLOps, Edge AI, Multimodal AI, Recommendation Systems, AI Agents, RAG

PROJECTS

CAFBrain: Multimodal LLM Platform (Agentic RAG | LLM) - [GitHub](#)

- Built an agentic RAG system that reduced grant proposal creation time from hours to under a minute, using Langchain and LangGraph with multi-step tool routing across 5000+ multimodal documents using FAISS vector db.

FSTChangeNet: Temporal Change Retrieval (Computer Vision | Multimodal) - [GitHub](#)

- Designed a dual-encoder architecture with custom frequency-spatial-temporal attention module and LoRA fine-tuned RemoteCLIP for vision-language alignment, achieving 64% Recall@10 on natural-language change queries for satellite images.

Scalable DBaaS for RideShare (Distributed Systems | Cloud Infra) - [Github](#)

- Designed a fault-tolerant Database-as-a-Service on AWS supporting 2000+ RPS with Zookeeper-based leader election, automatic failover, and read/write routing via RabbitMQ RPC, achieving zero-downtime recovery under node failure.

EDUCATION

Master of Science (M.S) in Machine Learning Aug 2024 - May 2026

University of Maryland, College Park

GPA: 3.8/4

Bachelor of Technology (B.Tech) in Computer Science Aug 2017 - May 2021

PES University, Bangalore

GPA: 3.6/4